

THEME D - Continuity & Change

Exam: Paper 1A (MCQ) · Paper 1B (Data) · Paper 2A (Short answer) · Paper 2B (Extended response)

WHAT'S IN THIS GUIDE

D1.1 DNA Replication — D1.2 Protein Synthesis — D1.3 Mutations & Gene Editing — D2.1 Cell & Nuclear Division — D2.3 Water Potential — D3.1 Reproduction — D3.2 Inheritance — D3.3 Homeostasis — D4.1 Natural Selection — D4.2 Stability & Change — D4.3 Climate Change

This theme asks: how is genetic information copied, expressed, and passed on — and how do populations and ecosystems change over time? It is the most content-heavy theme and includes genetics, inheritance, homeostasis, evolution, and environmental change.

High-frequency exam topics: DNA replication enzymes, transcription/translation, mutations, mitosis vs meiosis, water potential, Punnett squares (including sex linkage), blood glucose regulation, diabetes, natural selection (Endler's guppies), eutrophication, biomagnification, coral bleaching.

D1.1 — DNA Replication

Semi-conservative model · Meselson-Stahl · Enzymes · Leading/lagging strand · PCR · Gel electrophoresis

DNA replication ensures that every new cell receives a complete copy of the genetic information. It occurs in the nucleus before cell division (S phase of interphase). The process is semi-conservative — each new DNA molecule retains one original (template) strand.

KEY DEFINITIONS

Semi-conservative replication: each daughter DNA molecule consists of ONE original (parental) strand and ONE newly synthesised strand. The two strands of the parent DNA separate and each serves as a template.

Meselson-Stahl experiment (1958): provided definitive evidence for semi-conservative replication using heavy nitrogen (^{15}N) and density gradient centrifugation. After ONE generation in ^{14}N medium: all DNA was 'intermediate' density (one ^{15}N + one ^{14}N strand). After TWO generations: equal amounts of intermediate and light DNA. This pattern is only consistent with semi-conservative replication.

Enzyme	Function in DNA replication	Key detail
Helicase	Unwinds and separates (unzips) the two DNA strands by breaking hydrogen bonds between bases	Creates the replication fork. Works in both directions from the origin of replication.
Primase	Synthesises a SHORT RNA PRIMER — a short sequence of RNA complementary to the template strand	DNA polymerase cannot start a new chain — it can only extend an existing one. Primers provide the starting 3'—OH group.
DNA polymerase III (main)	Adds nucleotides (deoxyribonucleotides) to the 3' end of the growing strand — reads template 3'→5', synthesises 5'→3'	Can only synthesise in the 5'→3' direction. This creates the leading/lagging strand problem.
DNA polymerase I	Removes RNA primers and replaces them with DNA nucleotides	After the main strand is synthesised.
DNA ligase	Joins (seals) adjacent fragments of DNA by forming phosphodiester bonds — joins Okazaki fragments on lagging strand	Also joins RNA primer replacement DNA to adjacent strand. Forms covalent bond.
Topoisomerase	Relieves the tension (supercoiling) ahead of the replication fork as helicase unwinds the double helix	Without topoisomerase, the DNA would become too tightly coiled to unwind. Targeted by some antibiotics and anticancer drugs.

KEY DEFINITIONS

Leading strand: synthesised CONTINUOUSLY in the 5'→3' direction, moving TOWARD the replication fork — only needs one primer.

Lagging strand: synthesised DISCONTINUOUSLY in the 5'→3' direction, but moving AWAY from the fork — requires multiple primers and produces OKAZAKI FRAGMENTS (short fragments ~100-200 nucleotides). Ligase joins these fragments.

Origin of replication: specific sequence where replication begins. Eukaryotes have MULTIPLE origins of replication (allows faster replication of large chromosomes). Prokaryotes have ONE.

PROCESS: PCR (Polymerase Chain Reaction) — Amplifying DNA

1. PURPOSE: Amplify (make millions of copies of) a specific DNA sequence from a tiny sample. Used in forensics, diagnosis, paternity testing, evolutionary studies, COVID-19 testing.
2. DENATURATION (~95°C): Double-stranded DNA is heated → hydrogen bonds break → strands separate.
3. ANNEALING (~55–65°C): Temperature lowered → short synthetic DNA sequences (PRIMERS) bind to complementary sequences flanking the target region on each strand.
4. EXTENSION (~72°C): TAQ POLYMERASE (heat-stable DNA polymerase from *Thermus aquaticus* — a thermophile) extends from the primers, synthesising new DNA strands. Optimal temperature for Taq.
5. REPEAT: Each cycle doubles the amount of target DNA. After 30 cycles: ~10⁹ copies from a single molecule. The 3 steps repeat automatically in a thermal cycler.

KEY DEFINITIONS

Gel electrophoresis: technique to separate DNA fragments by SIZE. DNA is loaded into wells in an agarose gel. Electric current applied — DNA (negatively charged, due to phosphate groups) migrates TOWARD the positive electrode. SMALLER fragments travel FURTHER (less resistance from gel). Results in a banding pattern. DNA stained with ethidium bromide (or SYBR green) and visualised under UV light.

Uses of gel electrophoresis: DNA profiling/fingerprinting (forensics — compare band patterns between samples), paternity testing, determining PCR product size, separating restriction enzyme digestion products.

EXAM TIP

Enzyme functions: the most commonly examined part of D1.1. Know ALL 6 enzymes by name and precise function. Helicase ≠ DNA polymerase. Ligase joins — it does not add nucleotides.

Why Taq polymerase for PCR? It is HEAT-STABLE — from the thermophile *Thermus aquaticus*. Regular DNA polymerase would be denatured at the 95°C denaturation step.

Gel electrophoresis: smaller = faster = travels further from the well. Larger = slower = stays closer to well. DNA migrates toward POSITIVE electrode (DNA is negatively charged).

D1.2 — Protein Synthesis

Transcription · mRNA processing · Translation · Genetic code · Degeneracy · Universality

Protein synthesis — the process by which genetic information in DNA is used to build polypeptides — occurs in two main stages: transcription (DNA → mRNA, in the nucleus) and translation (mRNA → polypeptide, at ribosomes in the cytoplasm).

PROCESS: Transcription — DNA to mRNA (in the nucleus)

1. INITIATION: RNA polymerase binds to the PROMOTER region (specific DNA sequence upstream of the gene). The double helix unwinds at this point.
2. ELONGATION: RNA polymerase moves along the TEMPLATE STRAND (3'→5') reading it and synthesising a complementary mRNA strand in the 5'→3' direction. Note: in mRNA, URACIL (U) replaces THYMINE (T) as the complement of Adenine (A).
3. TERMINATION: RNA polymerase reaches the TERMINATOR sequence → transcription stops → pre-mRNA strand released.
4. PRE-mRNA PROCESSING (eukaryotes only): 5' cap (7-methylguanosine) added to 5' end. Poly-A tail (100-200 adenine nucleotides) added to 3' end. SPLICING: introns (non-coding sequences) are removed by the spliceosome; exons (coding sequences) are joined together. The mature mRNA exits the nucleus through nuclear pores.
5. Template strand vs coding strand: the TEMPLATE (antisense) strand is READ by RNA polymerase. The CODING (sense) strand has the same sequence as the mRNA (except T→U). Exam questions give you coding strand sequences — remember mRNA has same sequence as coding strand.

🌀 PROCESS: Translation — mRNA to Polypeptide (at ribosomes)

- 1. INITIATION:** Small ribosomal subunit binds to 5' end of mRNA and scans for START CODON (AUG — codes for methionine). Large subunit joins. First tRNA (carrying methionine, anticodon UAC) enters the P site.
- 2. ELONGATION — CODON RECOGNITION:** Next codon in A site is read. tRNA with complementary ANTICODON brings the correct amino acid to the A site.
- 3. ELONGATION — PEPTIDE BOND FORMATION:** Peptidyl transferase (part of rRNA — ribozyme activity) catalyses formation of a PEPTIDE BOND between the amino acid in the P site and the incoming amino acid in the A site.
- 4. ELONGATION — TRANSLOCATION:** Ribosome moves 3 nucleotides (one codon) along mRNA in 5'→3' direction. tRNA in A site moves to P site. Empty tRNA exits from E site. New codon exposed in A site. Repeat.
- 5. TERMINATION:** Ribosome reaches a STOP CODON (UAA, UAG, or UGA). No tRNA has an anticodon complementary to stop codons. RELEASE FACTOR protein binds → polypeptide released. Ribosome dissociates.

Property of genetic code	Meaning	Significance
Triplet (codon-based)	Each codon = 3 consecutive nucleotide bases on mRNA. Codes for one amino acid.	3 bases × 4 possible bases = 64 possible codons. 20 amino acids → code is more than sufficient.
Degenerate	Most amino acids are coded for by MORE THAN ONE codon (multiple codons → same amino acid). E.g. UUA, UUG, CUU, CUC, CUA, CUG all code for leucine.	Silent (synonymous) mutations possible — third base of codon often varies without changing the amino acid. Reduces effect of point mutations. NOT ambiguous (each codon codes for only ONE amino acid).
Non-overlapping	Each nucleotide belongs to only ONE codon — codons do not share bases.	Frame-shifts (insertions/deletions) are devastating because they shift ALL downstream codons.
Universal	The same codons code for the same amino acids in virtually ALL living organisms — from bacteria to humans.	Evidence for COMMON ANCESTRY. Enables recombinant DNA technology — a human gene inserted into bacteria produces the correct human protein.
Start codon	AUG — codes for methionine — ALWAYS the first codon translated.	Sets the reading frame for the entire mRNA. All polypeptides start with methionine (often removed post-translationally).
Stop codons	UAA, UAG, UGA — no corresponding amino acid. No tRNA complement.	Signal end of translation. Release factors recognise stop codons.

📌 EXAM TIP

Transcription: RNA polymerase reads the TEMPLATE strand 3'→5' and synthesises mRNA 5'→3'. The mRNA has the same sequence as the CODING strand (except U for T).

Introns vs exons: 'introns are OUT (removed), exons are EXpressed'. Splicing removes introns — only exons appear in mature mRNA.

Degeneracy ≠ ambiguity. Degenerate means multiple codons → one amino acid. Ambiguous would mean one codon → multiple amino acids (this does NOT happen).

D1.3 — Mutations & Gene Editing

Point mutations · Frameshift · Consequences · CRISPR-Cas9

Mutation type	Definition	Effect on protein	Example
Silent (synonymous) substitution	One base replaced by another, but the new codon codes for the SAME amino acid (degeneracy of code)	No change in amino acid sequence — no change in protein structure or function	UCU → UCC (both code for serine)
Missense (non-synonymous) substitution	One base replaced → codon codes for a DIFFERENT amino acid	Amino acid change → may alter protein folding → altered or lost function. Severity depends on position and chemical similarity of new amino acid.	Sickle cell anaemia: GAG → GUG in β-globin → glutamate replaced by valine → abnormal haemoglobin (HbS) → red blood cells sickle in low O ₂

Nonsense substitution	One base replaced → creates a PREMATURE STOP CODON	Translation terminates early → truncated (shortened) polypeptide → usually non-functional	UGG → UGA (tryptophan codon → stop codon) → short, non-functional protein
Insertion / Deletion (frameshift)	One or more bases inserted or deleted → shifts the reading frame	ALL codons DOWNSTREAM of the mutation are changed → completely different amino acid sequence from that point → usually non-functional protein	Insertion of one G: ...AGG-CUG... → ...AGG-GCU-G... → reading frame shifted for entire downstream sequence

KEY DEFINITIONS

Mutagen: any agent that increases the rate of mutation. Types: UV radiation (causes thymine dimers — adjacent thymines bond together, blocking replication), ionising radiation (X-rays, gamma rays — break DNA strands), chemical mutagens (alkylating agents, base analogues — alter bases or mimic bases causing mispairing).

Spontaneous mutations: errors in DNA replication that escape proofreading by DNA polymerase. Very rare (~1 per 10⁹ base pairs per replication).

PROCESS: CRISPR-Cas9 Gene Editing — How It Works

- 1. DESIGN:** A GUIDE RNA (gRNA) is designed — 20 nucleotides complementary to the TARGET DNA sequence to be edited. The gRNA is combined with the Cas9 protein to form a complex.
- 2. TARGETING:** The gRNA-Cas9 complex scans the genome, base-pairing with complementary DNA sequences. It is directed to the specific target location by the gRNA sequence.
- 3. CLEAVAGE:** Cas9 (a nuclease — DNA-cutting enzyme) makes a DOUBLE-STRAND BREAK (cuts both strands) at the target site.
- 4. REPAIR — NHEJ:** The cell's natural repair mechanism (Non-Homologous End Joining) re-joins the cut ends — this is ERROR-PRONE, often causing small insertions or deletions (indels) → disrupts the gene → GENE KNOCKOUT (useful for studying gene function).
- 5. REPAIR — HDR:** If a TEMPLATE DNA with the desired sequence is provided, Homology-Directed Repair uses it as a template → PRECISE EDITING of the sequence → can correct mutations or insert new sequences.
- 6. APPLICATIONS:** Disease treatment (correcting sickle cell anaemia, cystic fibrosis), crop improvement (disease resistance, yield), basic research (studying gene function).

CRISPR — Ethical Considerations

SOMATIC vs GERMLINE editing: Somatic cell editing (non-reproductive cells) affects only one individual — changes are not heritable. Germline editing (embryos, eggs, sperm) would be HERITABLE — passed to all future generations.

Germline editing: deeply controversial. Concerns: off-target effects (unintended cuts elsewhere in genome), unknown long-term consequences, designer babies (selecting for traits like intelligence or appearance), justice (who has access?), consent (future generations cannot consent).

The 2018 'He Jiankui case': Chinese scientist claimed to have edited human embryos (CCR5 gene) → implanted → twin girls born. International condemnation. Highlighted need for governance frameworks.

Off-target effects: Cas9 may cut at unintended sites with similar sequences → unintended mutations. Research into more precise Cas9 variants (base editors, prime editors) aims to address this.

EXAM TIP

Frameshift vs substitution: frameshift (insertion/deletion of 1 or 2 bases) disrupts ALL downstream codons — usually catastrophic. Insertion/deletion of 3 bases adds/removes ONE amino acid but keeps the rest of the reading frame intact.

CRISPR questions: know the GUIDE RNA role (directs Cas9 to target), Cas9 role (cuts DNA), and the TWO repair pathways (NHEJ = knockout, HDR = precise edit). Also know at least 2 ethical concerns.

D2.1 — Cell & Nuclear Division

Mitosis · Meiosis · Genetic variation · Non-disjunction · Down syndrome · Karyotypes

Cell division is fundamental to growth, repair, and reproduction. Mitosis produces genetically identical cells for growth and repair. Meiosis produces genetically unique gametes for sexual reproduction.

Phase of mitosis	What happens	Key visual features	Memory aid
Prophase	Chromatin condenses into visible chromosomes (each consisting of 2 sister chromatids joined at centromere). Spindle fibres form from centrioles (animals). Nuclear envelope breaks down.	Short thick chromosomes visible. Spindle fibres forming.	Pro = PROpare
Metaphase	Chromosomes align at the METAPHASE PLATE (equator of cell). Spindle fibres attach to centromeres of each chromosome.	Chromosomes lined up in the middle — clearest view of chromosome number.	Meta = Middle
Anaphase	Sister chromatids are separated — spindle fibres shorten and pull chromatids to OPPOSITE POLES. Each pole receives one copy of each chromosome.	V-shaped (or fish-tail shaped) chromatids being pulled apart. Cell elongates.	Ana = Apart
Telophase	Nuclear envelopes reform around each set of chromosomes. Chromosomes decondense back to chromatin. Spindle fibres disappear.	Two nuclei forming at poles. Cell beginning to divide.	Telo = Two forming
Cytokinesis	Cytoplasm divides. In animals: cleavage furrow (actin-myosin ring pinches). In plants: cell plate forms (Golgi vesicles fuse along equator → new cell wall).	Two separate daughter cells produced. Genetically IDENTICAL to parent cell and to each other.	Cyto = Cytoplasm division

Feature	Mitosis	Meiosis
Number of divisions	ONE division	TWO divisions (meiosis I + meiosis II)
Number of daughter cells	2	4
Ploidy of daughter cells	DIPLOID (2n) — same as parent	HAPLOID (n) — half parent
Genetic identity of daughter cells	Genetically IDENTICAL to parent (unless mutation)	Genetically UNIQUE — different from parent and from each other
Where it occurs	All dividing somatic (body) cells	Gonads only — ovaries (oogenesis) and testes (spermatogenesis)
Purpose	Growth, repair, asexual reproduction	Production of gametes (sex cells) for sexual reproduction
Homologous pairing	Does NOT occur	DOES occur in prophase I — bivalents form
Crossing over	Does NOT occur	DOES occur in prophase I between non-sister chromatids of homologous chromosomes — creates new allele combinations
Independent assortment	Does NOT apply	DOES occur at metaphase I — random orientation of bivalents creates variation

KEY DEFINITIONS

Homologous chromosomes: a pair of chromosomes (one from each parent) with the same gene loci in the same order. They may carry different alleles.

Crossing over (recombination): in prophase I of meiosis, homologous chromosomes pair up (form BIVALENTS). Non-sister chromatids exchange sections at points called CHIASMATA. Creates new combinations of alleles on chromosomes — increases genetic variation.

Independent assortment: at metaphase I, bivalents (homologous pairs) line up randomly at the equator. Each pair can orient in either of two ways. With 23 pairs in humans: 2^{23} possible combinations (~8 million) from independent assortment alone.

Non-disjunction: failure of homologous chromosomes (meiosis I) or sister chromatids (meiosis II or mitosis) to separate properly during cell division. Results in cells with wrong chromosome number (aneuploidy).

Down syndrome (Trisomy 21): most common human chromosomal abnormality. Results from non-disjunction of chromosome 21 (usually in meiosis I of oogenesis). Individual has 47 chromosomes instead of 46. Risk increases with maternal age.

EXAM TIP

Mitosis vs meiosis comparison: most important table in this sub-topic. Know ALL differences. Key ones examiners look for: number of divisions, haploid vs diploid, crossing over, genetic identity.

Sources of genetic variation in meiosis: (1) CROSSING OVER in prophase I, (2) INDEPENDENT ASSORTMENT of homologous pairs in metaphase I, (3) RANDOM FERTILISATION (any sperm can fertilise any egg).

Non-disjunction: can happen in meiosis I (both homologues go to same cell) or meiosis II (sister chromatids fail to separate). Also possible in mitosis (leads to cancer if tumour suppressor genes affected).

D2.3 — Water Potential

Ψ formula · Osmosis direction · Turgor · Plasmolysis · Medical applications

Water potential (Ψ) is the measure of the tendency of water to move from one place to another. Water always moves from regions of HIGHER water potential (less negative) to regions of LOWER water potential (more negative) — by osmosis through semi-permeable membranes.

KEY DEFINITIONS

Water potential (Ψ): measured in pressure units (kilopascals, kPa). PURE WATER has $\Psi = 0$ (the maximum possible). Any solute added LOWERS water potential (makes it more negative). Pressure increases water potential.

$\Psi = \Psi_s + \Psi_p$: total water potential = solute potential + pressure potential.

Solute potential (Ψ_s): ALWAYS NEGATIVE or zero. More dissolved solutes $\rightarrow$ more negative $\Psi_s \rightarrow$ lower water potential. Lowering Ψ_s drives water IN by osmosis (in plants under pressure).

Pressure potential (Ψ_p): USUALLY POSITIVE in plant cells (due to cell wall resisting expansion — turgor pressure). In animal cells and open systems: $\Psi_p = 0$.

Water movement: ALWAYS from HIGH Ψ (less negative) to LOW Ψ (more negative). 'Water moves toward lower water potential.'

Scenario	What happens	Explanation	Result
Plant cell in HYPOTONIC solution (more dilute than cell contents)	Water moves INTO cell by osmosis	Cell cytoplasm has lower Ψ (more negative) than external solution. Water moves from high Ψ (external) to low Ψ (cell).	Cell swells. Cell wall resists expansion — Ψ_p increases. Cell becomes TURGID. When Ψ inside = Ψ outside, net water movement stops.
Plant cell in HYPERTONIC solution (more concentrated than cell contents)	Water moves OUT OF cell by osmosis	External solution has lower Ψ (more negative) than cell. Water moves out from cell (high Ψ) to solution (low Ψ).	Cell loses water. Vacuole shrinks. PLASMOLYSIS: cell membrane pulls away from cell wall. $\Psi_p = 0$ at incipient plasmolysis.
Animal cell in HYPOTONIC solution	Water moves INTO cell	Cell contents have lower Ψ than external solution. Water enters.	Cell Lyses (bursts) — no cell wall to resist. Haemolysis of red blood cells.
Animal cell in HYPERTONIC solution	Water moves OUT OF cell	External solution has lower Ψ . Water leaves.	Cell CRENATES (shrinks and shrivels). Red blood cells crenate.
Animal cell in ISOTONIC solution (same concentration as cell)	No net movement of water	Ψ inside = Ψ outside. Equal movement in both directions.	Cell maintains normal shape and volume.

Water Potential — Medical and Practical Applications

IV fluids (intravenous drip): must be ISOTONIC to blood (~0.9% NaCl — 'normal saline' or 5% glucose). If hypotonic $\rightarrow$ red blood cells lyse (haemolysis). If hypertonic $\rightarrow$ red blood cells crenate and dehydrate. Isotonic maintains normal cell shape and volume.

Oral rehydration therapy: glucose + salt solution for diarrhoea/dehydration — isotonic composition maximises water absorption in the gut.

Food preservation: salt and sugar in high concentrations create hypertonic environment $\rightarrow$ water leaves bacteria/fungi by osmosis $\rightarrow$ dehydration $\rightarrow$ microorganism death (antimicrobial effect). Jam, pickles, salted fish.

Contractile vacuoles: organelles in freshwater unicellular organisms (e.g. Paramecium, Amoeba). Freshwater is HYPOTONIC $\rightarrow$ water constantly enters by osmosis. Contractile vacuole collects excess water and periodically EXPELS it by exocytosis — prevents lysis.

EXAM TIP

Water potential numbers: remember Ψ of pure water = 0. Adding solutes makes Ψ MORE NEGATIVE. Higher (less negative) $\Psi \rightarrow$ lower (more negative) Ψ . Water moves toward MORE CONCENTRATED solutions (lower Ψ).

Turgor pressure in plants: makes non-woody plants rigid (wilting when water-stressed = loss of turgor). Turgid cells have $\Psi_p > 0$. Plasmolysed cells have $\Psi_p = 0$.

D3.1 — Reproduction

Asexual vs sexual · Life cycle · Human reproduction · Plant reproduction · IVF

Feature	Asexual reproduction	Sexual reproduction
Offspring	Genetically IDENTICAL to parent (clones)	Genetically UNIQUE (variation from meiosis + fertilisation)
Number of parents	ONE parent	TWO parents (usually)
Gametes involved	No — mitosis only	YES — meiosis produces haploid gametes
Speed	Fast — can reproduce rapidly	Slower — finding mate, courtship, fertilisation
Advantage	Rapidly colonise favourable environment. All offspring can reproduce. No energy wasted on finding mate.	Genetic variation $\rightarrow$ better adaptation to changing environments $\rightarrow$ natural selection can act. Variation reduces spread of disease through population.
Examples	Binary fission (bacteria), budding (yeast, Hydra), vegetative propagation (runners in strawberries), clonal crops	Sexual reproduction in all animals, most plants, many fungi

Stage of human female reproduction	Key hormones	Events
Follicular phase (days 1–14)	FSH (from pituitary) $\rightarrow$ stimulates follicle growth. Follicle secretes increasing OESTROGEN.	Follicles develop in ovary. Oestrogen causes proliferation of uterine lining (endometrium). Rising oestrogen triggers LH surge.
Ovulation (day ~14)	LH SURGE from pituitary	Mature follicle ruptures $\rightarrow$ egg released from ovary into fallopian tube. Triggered by high LH.
Luteal phase (days 14–28)	LH stimulates corpus luteum $\rightarrow$ secretes PROGESTERONE + some oestrogen.	Corpus luteum (remains of follicle) produces progesterone $\rightarrow$ maintains uterine lining for possible implantation. If no fertilisation $\rightarrow$ corpus luteum degenerates $\rightarrow$ progesterone falls $\rightarrow$ menstruation.
If fertilisation occurs	hCG from embryo maintains corpus luteum	Implantation $\rightarrow$ placenta forms $\rightarrow$ hCG maintains progesterone production $\rightarrow$ no menstruation $\rightarrow$ pregnancy.

KEY DEFINITIONS

Acrosome reaction: sperm head contains the ACROSOME (membrane-bound vesicle of hydrolytic enzymes). When sperm reaches the zona pellucida (glycoprotein coat of egg), acrosome releases its enzymes $\rightarrow$ digest a path through zona pellucida $\rightarrow$ sperm fuses with egg membrane.

Cortical reaction: immediately after fertilisation, cortical granules beneath the egg membrane release their contents $\rightarrow$ zona pellucida hardens (becomes impenetrable) $\rightarrow$ POLYSPERMY BLOCK — prevents more than one sperm from fertilising the egg.

Double fertilisation (angiosperms/flowering plants): one sperm nucleus + egg $\rightarrow$ zygote (2n). Second sperm nucleus + two polar nuclei $\rightarrow$ endosperm (3n) — nutritive tissue for developing embryo. Unique to flowering plants.

IVF (in vitro fertilisation): (1) Ovarian stimulation (FSH injections — superovulation). (2) Egg collection (transvaginal ultrasound-guided needle aspiration). (3) Fertilisation in lab dish. (4) Embryo culture (3–5 days). (5) Embryo transfer to uterus. Uses: blocked fallopian tubes, low sperm count, genetic screening of embryos (PGD).

EXAM TIP

Sexual life cycle: DIPLOID organism → MEIOSIS → HAPLOID gametes → FERTILISATION → DIPLOID zygote → MITOSIS → DIPLOID organism. Know the ploidy at each stage.

Plant vs animal fertilisation: plants have DOUBLE fertilisation (unique). Animals have single fertilisation but polyspermy prevention mechanisms (cortical reaction).

D3.2 — Inheritance

Punnett squares · Codominance · Incomplete dominance · Sex linkage · Pedigrees · Polygenic inheritance

Inheritance is the passing of genetic information from parents to offspring. Mendel's laws (segregation and independent assortment) underpin classical genetics. Modern genetics adds understanding of codominance, sex linkage, polygenic traits, and molecular variation.

KEY DEFINITIONS

Gene: a sequence of DNA that codes for a specific protein (or functional RNA). Located at a specific locus (position) on a chromosome.

Allele: one of two or more alternative forms of a gene at the same locus. Arise through mutation.

Dominant allele: expressed in the phenotype in both homozygous (AA) AND heterozygous (Aa) states.

Recessive allele: expressed ONLY in the homozygous state (aa). Masked by the dominant allele in heterozygotes.

Codominance: BOTH alleles are expressed simultaneously in the heterozygote — neither is dominant. Example: ABO blood group (type AB — both I^A and I^B expressed). Roan cattle (red + white hairs both present).

Incomplete dominance: heterozygote shows an INTERMEDIATE phenotype between the two homozygotes. Example: red (RR) × white (WW) flowers → pink (RW) offspring. Neither allele is dominant.

Multiple alleles: a gene with MORE THAN TWO possible alleles in the population. Example: ABO blood group has 3 alleles (I^A , I^B , i) — any individual has 2 of these 3.

Polygenic inheritance: phenotype determined by contributions from MULTIPLE GENES. Each gene adds to or subtracts from the phenotype. Results in CONTINUOUS VARIATION (bell curve). Examples: height, skin colour, IQ.

SNP (Single Nucleotide Polymorphism): single nucleotide position in the genome where two or more alternative bases occur (each with frequency > 1%). Most common form of genetic variation in humans. Used in genome-wide association studies (GWAS) to identify disease risk genes.

ABO Blood Groups — Worked Example of Multiple Alleles + Codominance

Three alleles: I^A (produces A antigen), I^B (produces B antigen), i (no antigen). I^A and I^B are CODOMINANT to each other. Both I^A and I^B are DOMINANT over i.

Blood type A: $I^A I^A$ or $I^A i$. Blood type B: $I^B I^B$ or $I^B i$. Blood type AB: $I^A I^B$ (codominant). Blood type O: ii (homozygous recessive).

Cross example: Type A ($I^A i$) × Type B ($I^B i$) → offspring: $I^A I^B$ (AB), $I^A i$ (A), $I^B i$ (B), ii (O) → 1:1:1:1 ratio of AB:A:B:O.

Clinical importance: blood transfusion compatibility. Type O = universal donor. Type AB = universal recipient. Anti-A and anti-B antibodies in plasma cause agglutination if incompatible blood given.

Sex-Linked Inheritance

KEY DEFINITIONS

Sex determination: humans — females: XX, males: XY. The SRY gene on the Y chromosome triggers male development. X chromosome carries hundreds of genes; Y chromosome has very few.

X-linked traits: genes located on the X chromosome. Males (XY) have only ONE X chromosome — if they inherit a recessive allele on their X, they express it (no second X to mask it). Females (XX) can be carriers ($X^A X^a$) — they carry the recessive allele but do not show the phenotype.

X-linked recessive examples: colour blindness (red-green — OPN1LW/OPN1MW genes), haemophilia A (Factor VIII gene), Duchenne muscular dystrophy.

Carrier female notation: $X^H X_h$ (carries haemophilia allele — not affected). Affected male: $X_h Y$. Affected female: $X_h X_h$ (must be homozygous recessive — rare).

Pedigree analysis: squares = male, circles = female. Shaded = affected. Horizontal line = mating. Vertical line = offspring. Common questions: determine inheritance pattern (autosomal or X-linked? dominant or recessive?) from pedigree, predict offspring ratios.

Cross type	Expected offspring ratio	Example
Monohybrid — F1 heterozygous × heterozygous	3 dominant : 1 recessive (3:1)	$Aa \times Aa \rightarrow AA, Aa, Aa, aa \rightarrow 3A_ : 1aa$
Test cross — unknown genotype × homozygous recessive	1:1 if heterozygous. 1:0 if homozygous dominant	$Aa \times aa \rightarrow Aa, aa (1:1)$. $AA \times aa \rightarrow \text{all } Aa$.
Dihybrid — F1 double heterozygous × double heterozygous	9:3:3:1 (independent assortment)	$AaBb \times AaBb \rightarrow 9A_B_ : 3A_bb : 3aaB_ : 1aabb$
X-linked recessive — carrier female × normal male	½ sons affected, ½ daughters carrier	$X^H X_h \times X^H Y \rightarrow X^H X^H : X^H X_h : X^H Y : X_h Y$
Codominance — heterozygote × heterozygote	1:2:1 (not 3:1 — both alleles visible)	$I^A i \times I^A i$ — WRONG for codominance. $I^A I^B \times I^A I^B \rightarrow I^A I^A : I^A I^B : I^B I^B (1:2:1)$

EXAM TIP

Always show your Punnett square workings in the exam — do not just state the ratio. Show parental genotypes, gametes, and the completed grid. Identifying inheritance pattern from pedigree: autosomal recessive — appears in both sexes equally, often skips generations. X-linked recessive — mostly males affected, carrier females. Dominant — appears in every generation, affected individuals have at least one affected parent. PKU (phenylketonuria): autosomal recessive metabolic disorder. Cannot metabolise phenylalanine → accumulates → damages developing brain. Managed by low-phenylalanine diet. All newborns screened (heel prick test). Classic example of genotype-environment interaction — the diet (environment) prevents the phenotype (intellectual disability).

D3.3 — Homeostasis

Negative feedback · Blood glucose · Insulin/Glucagon · Diabetes · Thermoregulation

Homeostasis is the maintenance of a stable internal environment despite changes in external conditions. It is achieved through negative feedback mechanisms — deviations from the set point trigger corrective responses that restore the set point.

PROCESS: Blood Glucose Regulation — Negative Feedback

1. HIGH BLOOD GLUCOSE (after eating): Detected by β cells of the islets of Langerhans in the pancreas.
2. INSULIN SECRETED: β cells secrete INSULIN into the blood.
3. TARGET CELL RESPONSE: Insulin binds to receptors on liver and muscle cells. LIVER: increases glycogen synthesis (glucose → glycogen = glycogenesis). MUSCLE: increases glucose uptake. ADIPOSE: increases fat synthesis. All cells: increases glucose uptake from blood.
4. RESULT: Blood glucose falls back toward the set point (~90mg/100mL).
5. LOW BLOOD GLUCOSE (fasting, exercise): Detected by α cells of the islets of Langerhans.
6. GLUCAGON SECRETED: α cells secrete GLUCAGON into the blood.
7. TARGET CELL RESPONSE: Glucagon binds to liver cells. LIVER: increases glycogen breakdown to glucose (glycogenolysis). Also stimulates gluconeogenesis (synthesis of glucose from non-carbohydrate sources e.g. amino acids, glycerol).
8. RESULT: Blood glucose rises back toward the set point.

Feature	Type 1 Diabetes Mellitus	Type 2 Diabetes Mellitus
Cause	AUTOIMMUNE destruction of β cells of islets of Langerhans → no insulin production	Cells develop INSULIN RESISTANCE — insulin is produced but target cells no longer respond adequately. Progressive loss of β cell function.
Insulin production	None — β cells destroyed	Initially normal or high (compensating), eventually reduced
Onset	Usually childhood/adolescence (juvenile-onset)	Usually adulthood — strongly associated with obesity, sedentary lifestyle, diet high in refined carbohydrates

Treatment	INSULIN INJECTIONS (or pump) — essential for survival. Carbohydrate counting, blood glucose monitoring.	Lifestyle changes (diet, exercise, weight loss). Oral medications (metformin — reduces liver glucose output). Insulin injections if advanced.
Blood glucose	Very high if untreated (hyperglycaemia)	Elevated, but typically less extreme than uncontrolled Type 1
Complications (both types)	Retinopathy (eye damage), nephropathy (kidney damage), neuropathy (nerve damage), cardiovascular disease, poor wound healing	Same complications — related to chronic hyperglycaemia damaging blood vessels and nerves

Thermoregulation

KEY DEFINITIONS

Hypothalamus: the body's thermostat. Has thermoreceptors that detect blood temperature. Also receives input from peripheral thermoreceptors in the skin.

Set point: ~37°C in humans. Core body temperature maintained within narrow range (~36.1–37.2°C).

Temperature deviation	Response	Mechanism
TOO COLD — hypothermia risk	Vasoconstriction	Blood vessels near skin surface constrict → less blood flow to skin → less heat lost from surface
TOO COLD	Shivering	Rapid involuntary muscle contractions → generate heat through metabolic activity
TOO COLD	Piloerection ('goosebumps')	Arrector pili muscles contract → hair stands on end → traps insulating air layer (vestigial in humans, effective in furred animals)
TOO COLD	Increased metabolic rate	Adrenaline and thyroid hormone increase overall metabolism → more heat generated
TOO COLD	Behavioural responses	Curling up (reduces SA:V), putting on more clothing, moving to warmer location
TOO HOT — hyperthermia risk	Vasodilation	Blood vessels near skin surface dilate → more blood flow to skin → more heat radiated/lost from surface → skin flushes
TOO HOT	Sweating	Sweat glands secrete water → evaporation from skin surface removes heat (latent heat of vaporisation)
TOO HOT	Behavioural responses	Moving to shade, reducing activity, panting (in some animals — evaporation from respiratory surfaces)

EXAM TIP

Insulin vs glucagon: INSULIN lowers blood glucose (secreted by β cells when glucose HIGH). GLUCAGON raises blood glucose (secreted by α cells when glucose LOW). They are ANTAGONISTIC hormones — act in opposition.

Type 1 vs Type 2 diabetes: TYPE 1 = no insulin produced (autoimmune β cell destruction). TYPE 2 = insulin resistance (insulin produced but ineffective). Treatment differs: Type 1 requires insulin. Type 2 can be managed without insulin initially.

Negative feedback for thermoregulation: deviation detected by hypothalamus → corrective response activated → temperature returns to set point → receptors detect return to normal → response switched off.

D4.1 — Natural Selection

Variation · Selection pressure · Endler's guppies · Types of selection · Allele frequency change

Natural selection is the mechanism by which evolution occurs. Heritable variation + environmental selection pressures → change in allele frequencies over time. D4.1 examines natural selection in detail, including a key experimental demonstration.

🌀 PROCESS: Natural Selection — The Complete Mechanism

- 1. OVERPRODUCTION:** Organisms produce more offspring than can possibly survive — there is always competition for limited resources (food, water, territory, mates). This creates a 'struggle for existence'.
- 2. HERITABLE VARIATION:** Individuals within a population differ in heritable traits — variation arises from mutation and sexual reproduction (crossing over, independent assortment, random fertilisation). Not all variation is heritable — only genetic variation can be selected.
- 3. SELECTION PRESSURE:** Environmental factors (predators, disease, climate, food availability, competition) determine which individuals survive and reproduce. Organisms with traits that increase survival/reproductive success in the current environment leave more offspring.
- 4. DIFFERENTIAL REPRODUCTION:** Individuals with advantageous traits are more likely to survive to reproductive age AND leave more offspring. This is 'survival of the fittest' — where 'fitness' means **REPRODUCTIVE SUCCESS** in that specific environment.
- 5. INHERITANCE AND CHANGE:** Advantageous alleles are passed to offspring. Over many generations, the **FREQUENCY** of advantageous alleles **INCREASES** in the population, while unfavourable alleles decrease. The population **EVOLVES**.
- 6. RESULT:** Populations become increasingly well-adapted to their environment. Different environments → different selection pressures → populations diverge → eventually speciation.

Endler's Guppy Experiment (1980) — Natural Selection Demonstrated in Real Time

John Endler studied guppies (*Poecilia reticulata*) in rivers in Trinidad. Male guppies have coloured spots — colour patterns vary genetically within populations.

KEY OBSERVATION: Guppy colour patterns differed between pools based on predator community: **HIGH** predation sites (dangerous predators e.g. pike cichlid) → guppies were **DULL COLOURED** (camouflaged — hard to spot). **LOW** predation sites (mild predators e.g. killifish) → guppies were **BRIGHT COLOURED** (attractive to females — mate selection favours brightness when predation risk is low).

EXPERIMENTAL MANIPULATION: Endler moved dull-coloured guppies from high-predation pools to low-predation pools. After several generations: the introduced population **EVOLVED BRIGHTER COLOURING** — natural selection in the new (low-predation) environment favoured bright males (better mating success outweighed the small predation risk).

REVERSAL: When predators were introduced to low-predation pools: the population rapidly evolved **DULLER COLOURING** again.

SIGNIFICANCE: (1) Demonstrates natural selection **DIRECTLY** in controlled conditions. (2) Shows selection is **DIRECTIONAL** and environment-specific. (3) Evolution happened **RAPIDLY** — within observable timescale (years, not millions of years). (4) Shows the trade-off between sexual selection and natural selection.

Type of selection	Pattern of change	Example	Outcome
Stabilising selection	Selection AGAINST BOTH EXTREMES . Individuals near the MEAN have highest fitness. Reduces variation.	Human birth weight: very small or very large babies have higher mortality. Medium birth weight is optimal.	Population becomes more uniform around the mean. Maintains the status quo in stable environments.
Directional selection	Selection FAVOURS ONE EXTREME . Population mean shifts in one direction. Increases variation briefly then reduces it.	Antibiotic resistance in bacteria: resistant strains survive and reproduce. Guppy colour shift in Endler's experiment. Peppered moth (industrial melanism — dark moths favoured on soot-blackened trees).	Population evolves toward the favoured extreme. Can lead to adaptation to changed environment or speciation if continued.
Disruptive (diversifying) selection	Selection FAVOURS BOTH EXTREMES . Individuals near the mean have LOWEST fitness. Increases variation.	Darwin's finches on Santa Cruz: medium-beak individuals competed for both large and small seeds — both extremes outperformed intermediates. Oysters in polluted vs unpolluted areas.	Population splits into two distinct sub-groups. Can lead to SYMPATRIC SPECIATION over time.

📌 EXAM TIP

Endler's experiment: know the **SETUP** (guppies, different predator environments), **RESULT** (dull in high predation, bright in low predation), **MANIPULATION** (moved dull guppies to low predation → became brighter), and **SIGNIFICANCE** (demonstrates natural selection in real time, reversible).

Natural selection vs evolution: **NATURAL SELECTION** is the mechanism. **EVOLUTION** is the result (change in allele frequency over time). Selection acts on **PHENOTYPE**. Evolution changes **GENOTYPE** frequency.

D4.2 — Stability & Change

Ecosystem stability · Keystone species · Deforestation · Eutrophication · Biomagnification · Invasive species

KEY DEFINITIONS

Ecosystem stability: the ability of an ecosystem to maintain its structure and function over time, and to return to its original state after a disturbance. Two components: resistance (ability to withstand disturbance) and resilience (speed of recovery after disturbance). Generally, more DIVERSE ecosystems are more stable (more redundancy — other species can fill a lost species' role).

Keystone species: a species that has a disproportionately LARGE effect on its ecosystem relative to its abundance. Removing a keystone species causes dramatic changes ('trophic cascade'). Example: SEA OTTERS eat sea urchins → without otters, urchin population explodes → urchins overgraze kelp → kelp forest collapses → loss of habitat for hundreds of species.

Trophic cascade: indirect effects on lower trophic levels caused by changes at the top. Adding/removing top predators can dramatically alter entire ecosystem structure through indirect effects on all other trophic levels.

Environmental issue	Cause	Mechanism	Consequences	Example
Deforestation	Agriculture, logging, urban development, mining	Trees removed → habitat lost, soil exposed, roots gone	Loss of biodiversity (habitat destruction), increased CO ₂ (less photosynthesis, decomposition of wood), soil erosion (no roots), disruption of water cycle (less transpiration → less rainfall), displacement of indigenous communities	Amazon basin: ~17% deforested since 1970. 'Tipping point' concept — if too much deforested, remaining forest may not sustain itself and collapse to savanna
Eutrophication	Excess nutrients (nitrates and phosphates) from agricultural runoff (fertilisers), sewage, urban runoff entering water bodies	Nutrients → algal bloom → blocks light → aquatic plants die → algae die → decomposers (bacteria) respire → oxygen depleted → HYPOXIA/ANOXIA → fish and invertebrates suffocate	Dead zones in rivers, lakes, coastal waters. Loss of aquatic biodiversity. Contamination of drinking water.	Gulf of Mexico dead zone (Mississippi River runoff). Baltic Sea. Chesapeake Bay, USA.
Biomagnification	Persistent, fat-soluble toxins in the environment — DDT (pesticide), mercury, PCBs	Toxins ingested but not excreted — stored in fatty tissue. Each consumer CONCENTRATES the toxin from multiple prey → toxin concentration INCREASES up each trophic level	Apex predators most affected (eagles, dolphins, humans). Eggshell thinning in raptors (DDT → banned). Mercury in large fish → human neurological damage.	DDT levels: water 0.00005ppb → plankton 0.04ppb → small fish 0.5ppb → large fish 2ppb → osprey 25ppb (500,000× concentration). PCBs in orca whales.
Plastic pollution	Single-use plastics, microplastics from degradation, synthetic fibres from washing	Plastics accumulate in ocean gyres and on beaches. UV radiation breaks into MICROPLASTICS (<5mm). Ingested by marine organisms at all trophic levels.	Marine animal entanglement/ingestion. Microplastics in food chain → human consumption. Leaching of plastic additives (endocrine disruptors). Long persistence (centuries to degrade).	Great Pacific Garbage Patch. Microplastics found in Arctic ice, deep ocean sediments, human blood.

KEY DEFINITIONS

Sustainability: meeting the needs of the present generation without compromising the ability of future generations to meet their own needs (Brundtland Report, 1987). Applied to fishing (sustainable yields), forestry (replanting), agriculture (soil conservation), energy (renewables).

Invasive species effects on stability: introduced species may lack natural predators/parasites → population explodes → outcompetes native species → biodiversity loss → altered ecosystem function. Examples: water hyacinth blocking sunlight in African lakes, cane toads poisoning Australian predators.

EXAM TIP

Eutrophication sequence — know it in order: nutrients → algal bloom → blocks light → aquatic plants die → algae die → bacterial decomposition → O₂ depleted → fish die. Markers want the MECHANISM, not just the result.

Biomagnification vs bioaccumulation: bioaccumulation = build-up of toxin in ONE organism over time. Biomagnification = increasing concentration UP THE FOOD CHAIN. Both can be asked.

D4.3 — Climate Change

Global warming · Polar changes · Ocean changes · Coral bleaching · Wetlands · Upslope range shift

Climate change — driven by enhanced greenhouse gas emissions from human activity — is the most serious environmental challenge of the 21st century. It threatens biodiversity, ecosystem services, and human civilisation.

Climate change effect	Mechanism	Biological consequence	Feedback effect
Rising global temperature	Enhanced greenhouse effect: CO ₂ , CH ₄ , N ₂ O, H ₂ O vapour absorb outgoing infrared radiation → re-emit → warm atmosphere. Human emissions from fossil fuels, agriculture, deforestation.	Range shifts (poleward and upslope), phenological mismatch (flowers bloom before pollinators emerge), heat stress, increased wildfires, drought, disrupted breeding cycles	Permafrost thaw → methane release → further warming (positive feedback). Ice melt → less albedo → more heat absorbed (ice-albedo positive feedback).
Sea level rise	Thermal expansion of oceans + melting glaciers and ice sheets	Inundation of coastal habitats (mangroves, salt marshes, low-lying land), displacement of coastal human populations, loss of small island nations, saltwater intrusion into freshwater systems	None direct — but loss of coastal wetlands (which store carbon) → more CO ₂ → more warming
Ocean warming	Oceans absorb >90% of excess heat trapped by greenhouse gases	Coral bleaching (loss of zooxanthellae when temp rises even 1–2°C), poleward shift of fish populations, disrupted marine food webs, reduced productivity in some regions	Warmer water holds less O ₂ → deoxygenation → more anoxic zones
Ocean acidification	CO ₂ dissolves in seawater → carbonic acid → lowers pH (currently ~8.1, down from pre-industrial ~8.2)	Impairs shell/skeleton formation in calcifying organisms (molluscs, echinoderms, corals — CaCO ₃ dissolves in acidic water). Disrupts behaviour of fish (affects chemoreception).	Reduced phytoplankton → less CO ₂ fixation → more CO ₂ in atmosphere (positive feedback)
Disruption of thermohaline circulation	Melting Arctic ice → large volumes of cold, fresh water added → disrupts density-driven deep water formation → slows/disrupts Atlantic Meridional Overturning Circulation (AMOC)	Major climate shifts in Western Europe (currently warmed by AMOC). Disrupted marine nutrient cycling. Changes in rainfall patterns globally.	Potential 'tipping point' — once disrupted, may be difficult to restore
Upslope (altitudinal) range shifts	Species track suitable temperature conditions by moving to higher elevations as lowlands warm	Species at mountain tops have nowhere to go → local extinction. Timing of species interactions shifts (predator-prey, plant-pollinator mismatches). Endemic mountain species at greatest risk.	Loss of species diversity → ecosystem disruption

KEY DEFINITIONS

Keeling Curve: continuous measurement of atmospheric CO₂ at Mauna Loa Observatory, Hawaii, since 1958. Shows: annual oscillation (seasonal — Northern Hemisphere spring photosynthesis draws down CO₂ in summer, released in winter) AND long-term RISING TREND from ~315 ppm in 1958 to ~420 ppm in 2023. Definitive evidence of human-driven CO₂ increase.

Coral bleaching: when sea surface temperature rises even 1–2°C above seasonal maximum for several weeks, corals expel their symbiotic zooxanthellae (photosynthetic dinoflagellates living in coral tissue). Without zooxanthellae: coral turns white ('bleaches'), loses ~90% of its energy supply, and dies if thermal stress continues. Mass bleaching events have affected all major reef systems.

Wetlands restoration: wetlands (marshes, peat bogs, mangroves, fens) are among the most effective carbon sinks on Earth — STORE more carbon per unit area than tropical forests. They also filter water, reduce flooding, provide habitat, and support fisheries. Restoration of degraded wetlands is increasingly seen as a key climate mitigation AND adaptation strategy.

Mitigation vs adaptation: MITIGATION = actions to reduce greenhouse gas emissions or increase carbon sinks (renewable energy, reforestation, electric vehicles). ADAPTATION = adjusting to already-occurring climate change (sea walls, drought-resistant crops, early warning systems, managed retreat).

EXAM TIP

Coral bleaching mechanism: temperature rise → zooxanthellae expelled → coral white (bleached) → dies without energy supply. Always include the mechanism, not just the result.

Wetlands carbon storage: peat bogs store carbon as partially decomposed organic matter (anaerobic conditions prevent full decomposition).

Draining peatlands releases stored CO₂ — counterproductive. Restoring them sequesters carbon.

Positive feedback loops amplify change — be specific: Arctic ice melt → less reflective surface (lower albedo) → more solar energy absorbed → more warming → more ice melt. Permafrost thaw → methane (a powerful greenhouse gas) released → more warming.

LINKS TO OTHER TOPICS

D1.1 DNA Replication → A1.2 (nucleotides are the building blocks), C1.2 (ATP needed for replication), A2.2 (replication in nucleus)

D1.2 Protein Synthesis → A1.2 (nucleic acid structure), B1.2 (proteins produced), C2.2 (neurotransmitter receptors are proteins)

D1.3 Mutations → D3.2 (mutations create alleles for inheritance), D4.1 (mutations provide variation for natural selection), A4.1 (molecular evidence for evolution)

D2.1 Cell Division → B2.3 (stem cells divide by mitosis), D3.1 (meiosis produces gametes for reproduction), D3.2 (meiosis shuffles alleles for inheritance)

D2.3 Water Potential → B2.1 (osmosis), B3.2 (water uptake in plants), B3.1 (guard cell mechanism)

D3.2 Inheritance → A3.1 (species concept, karyotyping), D4.1 (natural selection acts on heritable traits), D1.3 (mutations create new alleles)

D3.3 Homeostasis → C3.1 (negative feedback), C2.2 (hypothalamus uses neural signals)

D4.1-D4.3 Evolution & Environment → A4.1 (speciation), A4.2 (conservation), C4.1-C4.2 (ecology and ecosystems)

THEME D — EXAM READINESS CHECKLIST

D1.1: DNA REPLICATION

- ☒ Semi-conservative replication — Meselson-Stahl evidence
- ☒ All 6 enzyme names and functions
- ☒ Leading vs lagging strand (and why)
- ☒ PCR — 3 steps with temperatures and enzymes
- ☒ Gel electrophoresis — what separates what and how

D1.2: PROTEIN SYNTHESIS

- ☒ Transcription steps — template strand, RNA polymerase, pre-mRNA processing
- ☒ Translation steps — initiation, elongation (codon-anticodon, peptide bond), termination
- ☒ Genetic code properties — degenerate, non-overlapping, universal, triplet
- ☒ Start and stop codons

D1.3: MUTATIONS & GENE EDITING

- ☒ 4 mutation types — silent, missense, nonsense, frameshift
- ☒ Why frameshifts are most severe
- ☒ CRISPR — guide RNA role, Cas9 role, NHEJ vs HDR outcomes
- ☒ 2+ ethical concerns about CRISPR

D2.1: CELL & NUCLEAR DIVISION

- ☒ 4 phases of mitosis + cytokinesis — what happens in each
- ☒ Mitosis vs meiosis — 9+ differences
- ☒ 3 sources of genetic variation from meiosis
- ☒ Non-disjunction and Down syndrome
- ☒ How crossing over creates variation

D2.3: WATER POTENTIAL

- ☒ Water potential formula ($\Psi = \Psi_s + \Psi_p$)
- ☒ Water moves from high to low Ψ
- ☒ Turgor vs plasmolysis in plants
- ☒ Lysis vs crenation in animal cells
- ☒ Why IV fluids must be isotonic

D3.1: REPRODUCTION

- ☒ Asexual vs sexual — 3+ differences
- ☒ Sexual life cycle — ploidy at each stage
- ☒ Menstrual cycle — hormones and stages
- ☒ Double fertilisation in plants
- ☒ IVF steps

D3.2: INHERITANCE

- ☒ Codominance vs incomplete dominance — examples
- ☒ ABO blood group cross (3 alleles)
- ☒ Sex linkage — why males more often affected
- ☒ Punnett squares for all cross types
- ☒ Polygenic inheritance and continuous variation
- ☒ Pedigree interpretation

D3.3: HOMEOSTASIS

- ☒ Negative feedback mechanism — all steps
- ☒ Insulin vs glucagon — which cell secretes, what it does
- ☒ Type 1 vs Type 2 diabetes — cause, treatment
- ☒ Thermoregulation — 4 responses to cold and heat
- ☒ Role of hypothalamus

D4.1: NATURAL SELECTION

- ☒ 5 steps of natural selection
- ☒ Endler's guppy experiment — setup, result, significance
- ☒ Stabilising vs directional vs disruptive selection — example each
- ☒ What changes — allele frequency

D4.2: STABILITY & CHANGE

- ☒ Keystone species — example and effect of removal
- ☒ Eutrophication — 5-step mechanism
- ☒ Biomagnification — mechanism and example
- ☒ Deforestation — 4 consequences
- ☒ Trophic cascade

D4.3: CLIMATE CHANGE

- ☒ Keeling Curve — what it shows, significance
- ☒ Coral bleaching — mechanism
- ☒ Ocean acidification — $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{carbonic acid} \rightarrow \text{CaCO}_3$ dissolution
- ☒ Upslope range shifts — why mountain species are at risk
- ☒ Wetlands — carbon storage and restoration value
- ☒ Mitigation vs adaptation